

Experiment Configuration Values

July 6, 2026

Uplink

System and Scenario Values

Variable	Value
Users (K)	3
Receive antennas per user ($N_{r,k}$)	(16, 16, 16)
Transmit antennas per user ($N_{t,k}$)	(4, 4, 4)
Blocklength limits (T_k)	(50, 60, 80)
Configured B for payload completion (B_k)	(1000, 2000, 3000)
Power (P_k)	(30, 30, 30)
SNR (dB)	(8, 8, 8)
Symbol rate ($f_{s,k}$) (symbols/s)	(15000, 15000, 15000)
Error probability (ε_k)	(10^{-10} , 10^{-10} , 10^{-10})

Optimization and Training Values

Variable	Value
Initial rate-constraint multiplier ($\lambda_{k,\ell}^{(\text{UL},r)}$)	0.1
Initial power-constraint multiplier ($\lambda_{k,\ell}^{(\text{UL},p)}$)	0.01
Maximum convergence epochs	120
Network learning rate	0.01
Rate-constraint learning rate	0.01
Power-constraint learning rate	0.001
Augmented-Lagrangian rho for rate constraint*	1.0
Augmented-Lagrangian rho for power constraint*	1.0
Print interval (epochs)	1
KKT primal tolerance (r_p)	10^{-5}
KKT complementarity tolerance (r_c)	10^{-5}
KKT stationarity tolerance (r_s)	10^{-3}
Reduced- n logging interval	10
Testing seed	3
Number of training seeds	3
Training maximum epochs	100
Fallback target bits	1
Coarse n -step for training	5
Full-block rollout weight*	1.0
Tail-feasible rollout weight*	1.0
Tail-frontier rollout weight*	1.5
Maximum total blocks	256
Minimum n value	1
n -search step	1

Downlink

System and Scenario Values

Variable	Value
Users (K)	3
Base-station antennas (N_b)	16
Receive antennas per user ($N_{r,k}$)	(4, 4, 4)
Blocklength limits (T_k)	(50, 60, 80)
Configured B for payload completion (B_k)	(1000, 2000, 3000)
Power (P_k)	(30, 30, 30)
SNR (dB)	(8, 8, 8)
Symbol rate ($f_{s,k}$) (symbols/s)	(15000, 15000, 15000)
Error probability (ε_k)	(10^{-10} , 10^{-10} , 10^{-10})
Initial bits per symbol	(4, 4, 4)

Optimization and Training Values

Variable	Value
Maximum convergence epochs	120
Print interval (epochs)	5
User update steps per epoch	1
User update learning rate	0.001
Initial rate-constraint multiplier ($\lambda_{k,\ell}^{(\text{DL},r)}$)	0.1
Initial power-constraint multiplier ($\lambda_\ell^{(\text{DL},p)}$)	0.01
Rate-constraint learning rate	0.01
Power-constraint learning rate	0.01
Augmented-Lagrangian rho for rate constraint*	1.0
Augmented-Lagrangian rho for power constraint*	1.0
KKT primal tolerance (r_p)	10^{-5}
KKT complementarity tolerance (r_c)	10^{-5}
KKT stationarity tolerance (r_s)	10^{-3}
Testing seed	3
Number of training seeds	3
Training maximum epochs	120
Fallback target bits	1
Coarse n -step for training	5
Full-block rollout weight*	1.0
Tail-feasible rollout weight*	1.0
Tail-frontier rollout weight*	1.5
Remaining-bits weight power	1.0
Minimum user weight	0.25
Network-rate weight*	0.15
Latency-penalty weight	0.5
Maximum total blocks	256
Minimum n value	1
n -search step	1

* Configured numerically but inactive under the reported settings.